Simulating Wildfire Mitigation: A Multi-Agent Approach to Community Behavior

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Appendix

Interviewee Demographics

Category	Group	% of Sample (n=20)
Compliance Status	Compliant	55.00
	Partially Compliant	36.00
	N/A	9.09
Insurance Status	Retained	55.00
	Non-Renewed	27.00
	N/A	18.00

Table 1: Summary of compliance and insurance status for the 20 interviewed homeowners.

Consent Statement. All 20 homeowners interviewed for this study were informed of the research purpose and voluntarily agreed to participate.

Agent Memory Seed Importance Scoring

Memory Seed Importance Scoring (1–10):

- 9–10 = defining life/property event
- 7–8 = significant action or strong emotional reaction
- 5–6 = meaningful conversation or decision about mitigation
- 3–4 = routine observation or minor step
- 1–2 = passing remark or peripheral detail

Simulation Judge Prompts

Each judgment scores a single dimension in an independent call. Behavioral plausibility and intervention responsiveness are shown the resident's objective circumstances but not the seed narrative; persona consistency is shown the seed narrative and memory seeds but not the simulation events. The judge (`openai/gpt-5.4`, temperature 0) writes its reasoning before assigning an integer score from 1 to 5 and returns JSON.

Behavioral Plausibility

```
You are evaluating ONE dimension: behavioral
    plausibility. You are shown this resident's
    objective circumstances, but not their personality,
    attitudes, or history. Judge whether the reasoning
    makes sense for someone in these circumstances. Do
    not judge whether it sounds like a particular person
    .

Judge the resident's REASONING. The decision is shown as
     context: you are assessing whether the reasoning
    that produced it makes sense for someone in these
    circumstances.

BEHAVIORAL PLAUSIBILITY: How plausible, if at all, is
    this reasoning for a homeowner facing this situation
    ?

1 = Not plausible. Incoherent, contradicts basic facts
    of the situation, or no homeowner would reason this
    way.
2 = Barely plausible. The reasoning is followable but
    rests on a clear misunderstanding of the situation
    or makes a leap no homeowner would make.
3 = Reasonable but generic. Sensible, but the reasoning
    could apply to almost anyone facing this situation.
4 = Reasonable and situationally specific. The reasoning
     engages concrete features of this situation (costs,
     timing, constraints).
5 = Nuanced situational understanding. Weighs competing
    pressures or tradeoffs specific to this situation in
     a way that shows a meaningful grasp of what is at
    stake.

Rate on an integer scale of 1-5. Use whole numbers only.
     Write your reasoning first, then your score.
```

Persona Consistency

```
You are an expert evaluator assessing resident decisions
     in a wildfire mitigation study. Read the seed
    narrative carefully to understand this resident's
    actual role and constraints before scoring.

You are evaluating ONE dimension: persona consistency.
    You are shown the resident's seed narrative and
    memory seeds, but not the events of the simulation.
    Judge only faithfulness to the seed.

PERSONA CONSISTENCY: How different, if at all, would
    this response be if the agent had a different seed
    narrative? Note: distinctive communication style,
    tone, and framing count as persona signals.
```

```
1 = Contradicts the seed personality or key memory seeds
    .
2 = No contradiction, but off. Nothing conflicts
    outright, but tone, priorities, or framing don't fit
     this agent.
3 = Consistent with a generic version of this persona
    type but lacking in nuance, for example, the agent's
     specific history, experiences, or distinctive voice
    .
4 = Reflects at least one detail unique to this agent,
    for example, a specific cost, person, place, opinion
    , characteristic tone, or prior experience from the
    seed narrative or memory seeds.
5 = Clearly reflects details unique to this agent:
    specific costs, people or places, distinctive
    opinions, characteristic tone, or prior experiences
    from the seed narrative or memory seeds.

Rate on an integer scale of 1-5. Use whole numbers only.
     Write your reasoning first, then your score.
```

Intervention Responsiveness

```
You are an expert evaluator assessing resident decisions
     in a wildfire mitigation study.

You are evaluating ONE dimension: intervention
    responsiveness. You are shown this resident's
    objective circumstances but not their personality,
    attitudes, or history. Judge only how specifically
    the response engages the content of the intervention
    .

INTERVENTION RESPONSIVENESS: How specifically, if at all
    , did the agent engage with the content of this
    intervention?

1 = Ignored the intervention or gave a fully generic
    response.
2 = Acknowledged the intervention happened, but the
    response could apply to multiple different
    interventions.
3 = Acknowledged the event and responded appropriately
    but did not engage with its specific details.
4 = Engaged with specifics. Referenced concrete details
    from this intervention, for example, deadlines,
    requirements, named parties, or amounts.
5 = Acted on specifics. The response turned on
    particular details of this intervention; a different
     deadline, amount, or requirement would have
    produced a different decision.

Rate on an integer scale of 1-5. Use whole numbers only.
     Write your reasoning first, then your score.
```

All three prompts instruct the judge to output JSON only, as a dictionary with keys `reasoning` (a string) and `score` (an integer from 1 to 5).

Detailed Agent Tuning and Evaluation

Parameter	Definition
Top-K	How many memories to pass to agent prompt at decision time (5/8/12)
Recency Weight (α)	How much to weight how recently a memory was created. Memories from earlier simulation days score lower using exponential decay
Importance Score (β)	How much to weight the importance score that a memory has
Relevance Weight (γ)	How much to weight the semantic similarity between the intervention and each memory
Retrieval Mode	Which algorithm computes the semantic similarity between intervention and each memory. Dense: HuggingFace embeddings + cosine similarity; Sparse: keyword matching; Hybrid: blend of both, controlled by sparse_weight

Table 2: Retrieval hyperparameter definitions.

Top-K	Recency Weight	Importance Weight	Relevance Weight	Retrieval Mode
8	1.0	1.0	1.0	Dense
5	1.0	1.0	1.0	Dense
12	1.0	1.0	1.0	Dense
8	0.5	1.0	2.0	Dense
8	0.5	1.0	3.0	Dense
8	1.0	2.0	1.0	Dense
8	2.0	1.0	1.0	Dense
8	1.0	1.0	1.0	Sparse
8	1.0	1.0	1.0	Hybrid

Table 3: Retrieval configurations tested.

Memory Retrieval Hyperparameter Tuning and Evaluation:

Configuration	Insight	Accuracy	Combined
Threshold=50, questions=3	7/10	7/10	7/10
Threshold=50, questions=5	7/10	6/10	6.5/10
Threshold=100, questions=3	7/10	6/10	6.5/10
Threshold=150, questions=3	6/10	6/10	6.0/10

Table 4: Reflection configurations tested.

Reflection Hyperparameter Tuning and Evaluation:

Detailed Simulation Results

Full Simulation Scores by Agent. Each cell = overall score (mean of BP, PC, IR) from the full-simulation judge, one call per agent evaluating the complete 60-day trajectory. Shows whether ablation effects are consistent across agents or driven by specific profiles.

Condition	Laura	Linda	Walter	Margaret	Miriam Voss
Baseline	4.62	4.51	4.38	4.46	4.67
Ablation 1 (No Reflection)	4.63	4.47	4.47	4.49	4.63
Ablation 2 (No Memory or Reflection)	4.51	4.51	4.60	4.47	4.63
Budget	4.53	4.49	4.54	4.53	4.46

Table 5: Overall score (mean of BP, PC, IR) by agent and condition, from the independent per-intervention judge (`GPT-5.4`, temperature 0, five scoring passes averaged).

Baseline Variant: Scoring by Criteria by Agent. Full system only (Sonnet 4.6, memory + retrieval + reflection). Each value is the exact judge score for that agent × intervention, not averaged across conditions.

Criterion	Laura	Linda	Walter	Margaret	Miriam Voss
Behavioral Plausibility	4.67	4.37	4.37	4.27	4.37
Persona Consistency	4.47	4.37	4.30	4.40	4.87
Intervention Responsiveness	4.73	4.80	4.47	4.70	4.77
Mean	4.62	4.51	4.38	4.46	4.67

Table 6: Baseline condition: scoring by criterion and agent, from the independent per-intervention judge (`GPT-5.4`, five passes averaged). Each value is averaged over the agent's six interventions.

Baseline Variant: Scoring by Intervention by Agent.

Behavioral Plausibility					
Intervention	Laura	Linda	Walter	Margaret	Miriam Voss
Day 1: Firewise ordinance notification	4.60	4.20	4.20	4.00	4.00
Day 12: Defensible space inspection	4.80	4.80	4.00	4.20	4.80
Day 24: Neighbor pressure	4.60	4.60	4.40	4.40	4.40
Day 36: Firewise community campaign	4.20	4.00	4.40	4.00	4.20
Day 48: Insurance non-renewal notice	4.80	4.40	5.00	4.80	4.80
Day 55: Resource assistance email	5.00	4.20	4.20	4.20	4.00
Mean	4.67	4.37	4.37	4.27	4.37

Table 7: Behavioral Plausibility by intervention and agent, Baseline condition (independent per-intervention `GPT-5.4` judge, five passes averaged).

Persona Consistency					
Intervention	Laura	Linda	Walter	Margaret	Miriam Voss
Day 1: Firewise ordinance notification	4.40	4.60	5.00	4.60	5.00
Day 12: Defensible space inspection	4.60	4.40	4.20	4.60	5.00
Day 24: Neighbor pressure	4.20	3.80	3.80	4.00	4.40
Day 36: Firewise community campaign	4.60	4.40	4.00	4.40	5.00
Day 48: Insurance non-renewal notice	4.20	4.60	4.40	4.00	5.00
Day 55: Resource assistance email	4.80	4.40	4.40	4.80	4.80
Mean	4.47	4.37	4.30	4.40	4.87

Table 8: Persona Consistency by intervention and agent, Baseline condition (independent per-intervention `GPT-5.4` judge, five passes averaged).

Intervention Responsiveness					
Intervention	Laura	Linda	Walter	Margaret	Miriam Voss
Day 1: Firewise ordinance notification	4.00	4.80	4.60	5.00	5.00
Day 12: Defensible space inspection	5.00	5.00	4.80	5.00	5.00
Day 24: Neighbor pressure	4.80	4.80	4.00	4.60	5.00
Day 36: Firewise community campaign	5.00	4.40	4.20	4.00	4.60
Day 48: Insurance non-renewal notice	5.00	5.00	5.00	5.00	4.40
Day 55: Resource assistance email	4.60	4.80	4.20	4.60	4.60
Mean	4.73	4.80	4.47	4.70	4.77

Table 9: Intervention Responsiveness by intervention and agent, Baseline condition (independent per-intervention `GPT-5.4` judge, five passes averaged).

Code Availability

The code is available at https://anonymous.4open.science/r/wildfire-agent-simulation/README.md.